

CS470 2026 Spring Final Prep Note

(release date: 26/06/11)

Please review all "question slides" as well.

If you have limited time, you may want to review the following points first.

(WARNING!) However, the exam questions will not be limited to the points below.

The exam structure

1. T/F Questions 1~2 Sets
2. Problem sets (algorithms, short answers, discussions): 2~4 sets
3. (new) Open/discussion questions: several subproblems or at most 1 set.

"MDPs are prerequisites for RL"

10. Reinforcement Learning -I

- Definition/Basic idea
- RL <> MDP (difference)
- **Model-based learning**
- Model-free learning
 - Passive RL
 - ♦ **Direct Evaluation**
- **Temporal Difference Learning**
 - Active RL: **Exploration vs. Exploitation**
- **Q-Learning**

11. Reinforcement Learning -II

- **Q-Learning**
 - **Exploration vs. Exploitation**
 - **e-greedy**
 - **Exploration function**
 - regret
- **Approximate Q-Learning**
 - **Feature-based representation**
 - **Linear value function**
 - Policy search

12. Reinforcement Learning - III

- Video understanding (Advanced)
 - LongVU (Advanced)
 - VideoQA / EMNLP23 (Advanced)
 - **REFT**
 - **PPO**
 - **GRPO**
 - DeepVideo-R1 (Advanced)

- VidChain (Advanced)
- Magma (Advanced)

13. LLM-based Agents and Prompt Optimization

- **Prompt Engineering**
 - Emergent abilities, why do we observe it? when can we expect more?
 - In-context learning
 - Chain-of-thought, Few-shot CoT, Zero-shot CoT
- **Soft Prompt Learning**
 - CLIP
 - CoOp
 - ProMetaR (advanced)
 - RPO
 - DAPT
- **LLM-based Hard Prompt Optimization**
 - TextGrad
 - Genetic algorithm
 - GEPA
 - PRESTO (advanced)
- System Prompts for LLM Agent System (**Video Lecture**)
 - CAMEL
 - MetaSPO
 - SPRIG
- Prompts for Multi-Agent System (MAS)
 - Anthropic's Multi-Agent Research System

14. MLP

- Perceptron
- Linear/FC/Dense Layer
- Activation functions and their necessity, pros, and cons
 - softmax, sigmoid, tanh, ReLU, LeakyReLU, ELU, etc.
- Num. parameters, the dimensionality of variables/params
- (Q) Why do neural networks need non-linear activation functions?
- [Mini-lab] Training NNs on the web
 - Hyperparameters: learning rate, activation functions, decision boundaries, (regularization)
 - Discussions/comparison

15. Backpropagation

- Backpropagation
 - Computational graph
 - Chain rule
 - upstream/local/downstream gradients,
 - Quiz
 - Simplification
 - Why does the training mode require more memory than the eval (inference) mode?
 - (Q) Vanishing and exploding gradient problems?
 - (Q) How does the choice of activation function affect gradient flow?

16. Introduction to Scaling Laws

- Motivation
 - **Definition**
 - Power-law from, linear in a log-log space
 - History Before Scaling Laws
 - Why do we observe a power law?
 - Power-Law Scaling of MSE
 - Power-Law Scaling & Empirical Results
 - Feed-forward ratio, Aspect ratio, Attention Head Dimension
 - Non-embedding Parameters
 - Architecture: Transformer vs. LSTM
 - Transfer Learning
 - **Compute-Optimal Model Behavior**
 - **Limitation of Power-Law Scaling**
 - **Small data, power-law, irreducible error regions**
 - **(Q) Which of the above regions are not covered in our scaling law equations? (Q) Can we incorporate it into the equation?**
- (Q) Given Graph, discussion referring important points in graphs**
- (Q) Data quality?**

17. Intermediate Scaling Laws

- Chinchilla
 - scaling exponents
 - ♦ Kaplan's analysis with fixed LR schedule
 - ♦ $N:D \sim 0.73, 0.27 \rightarrow 1:1$
 - Approach 1. Fix model sizes N and vary number of training tokens D
 - **Approach 2: IsoFLOP profiles (Vary model size for fixed compute budget)**
 - Approach 3: Fitting a parametric loss function
 - ♦ Optimal solution (derivation)
- **Data Constraints in Scaling Law**
 - **Scaling Law of Data Repetition (modified equation)**
 - **Scaling Law of Data Repetition & Composition (multiple curves and optimal strategy)**
- **Hyperparameters in Scaling Law**
 - Architecture
 - Optimizer
 - Aspect ratio
 - batch size

(Limitations of Scaling Laws)

(Q) Do loss-based scaling laws accurately predict emergent capabilities like reasoning or planning?

(Q) How can we incorporate explicit data quality and diversity metrics into a unified scaling law?

(Q) How scaling laws explicitly predict optimal hyperparameters using minimal small-scale trials?